

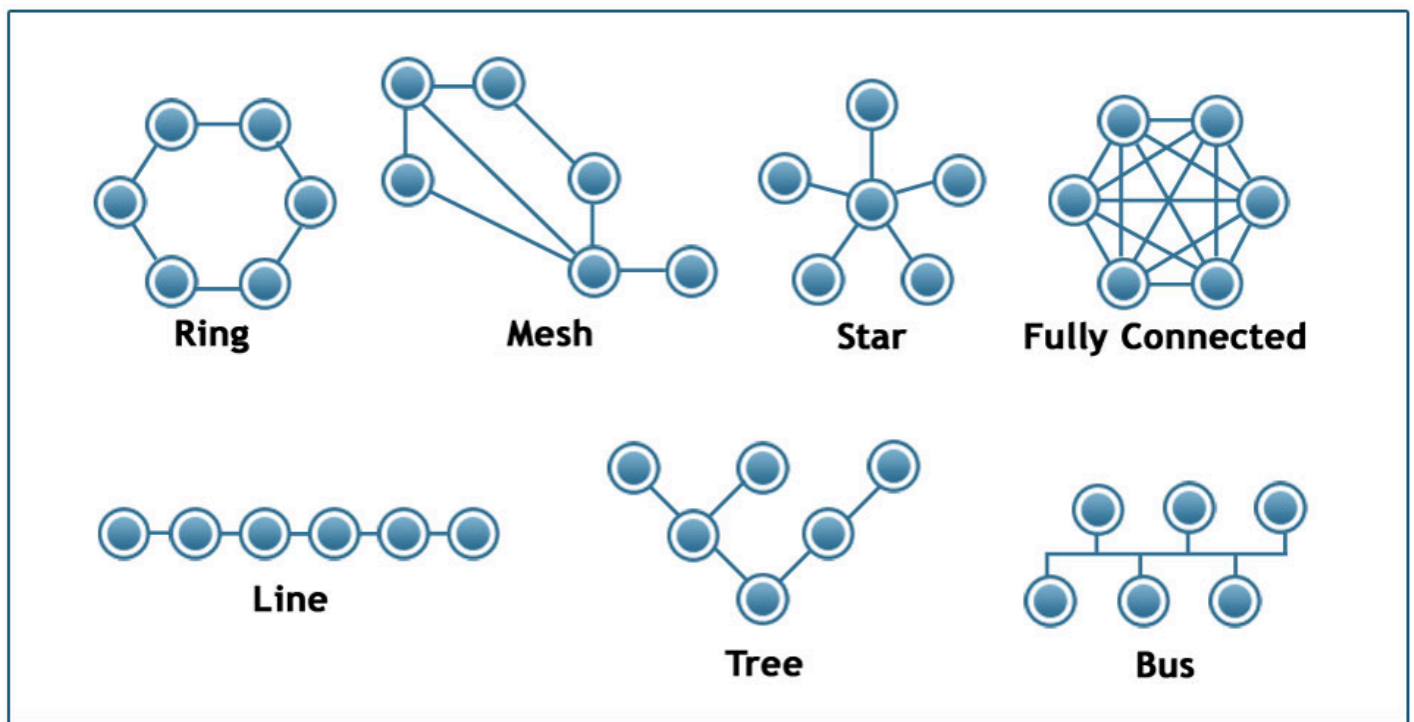
L1: The Main Premise

We can now state the main idea, the main premise, of network science:

The network architecture of a system provides valuable information about the system's function, capabilities, resilience, evolution, etc.

In other words, even if we don't know every little detail about a system and its components, simply knowing the map or "wiring diagram" that shows how the different system components are interconnected provides sufficient information to answer a lot of important questions about that system.

Or, if our goal is to design a new system (rather than analyze an existing system), network science suggests that we should first start from its network representation, and only when that is completely done, move to lower-level design and implementation.



Above, is an example to illustrate the previous point. (Image Source: techiereader.com → <https://techiereader.com/category/networking/>.)

Even if you know nothing about the underlying system, what would you say about its efficiency and resilience under each of the following architectures?

Suppose that we are to design a communication system of some sort that will interconnect 6 sites. The first question is: what should be the network architecture? This figure shows several options. For example, the Ring architecture provides two disjoint paths between every pair of nodes. The Line, Tree, and Star architectures require the fewest number of links but they are highly vulnerable when certain nodes or edges fail. The Fully Connected architecture requires the highest number of links but it also provides the most direct (*and typically faster*) and resilient communication. The Mesh architecture provides a trade-off between all previous properties.